

Features and User Journey

How a new user is onboarded, what gets configured, and what each person can do in the console.

Version 1.0 · 21 September 2026 · Satellite mission control console and its ground-segment services

1. What this document covers

This is a working guide to the product as it stands: the steps to bring a new team onto it, the screens each role uses, and the flows that matter most in daily operations. It deliberately leaves out system architecture.

Status labels. **Built** = works in the console today. **Demo data** = the screen works, but the numbers behind it are simulated because the service that would supply them does not exist yet. **Planned** = designed, not built.

2. Onboarding at a glance

Onboarding has two halves. Administrators set the workspace up once: who the tenant is, which stations and satellites exist, what the telemetry means, who may do what. After that, each new person goes through a short first-use path of sign-in, role, shift confirmation and personalisation.

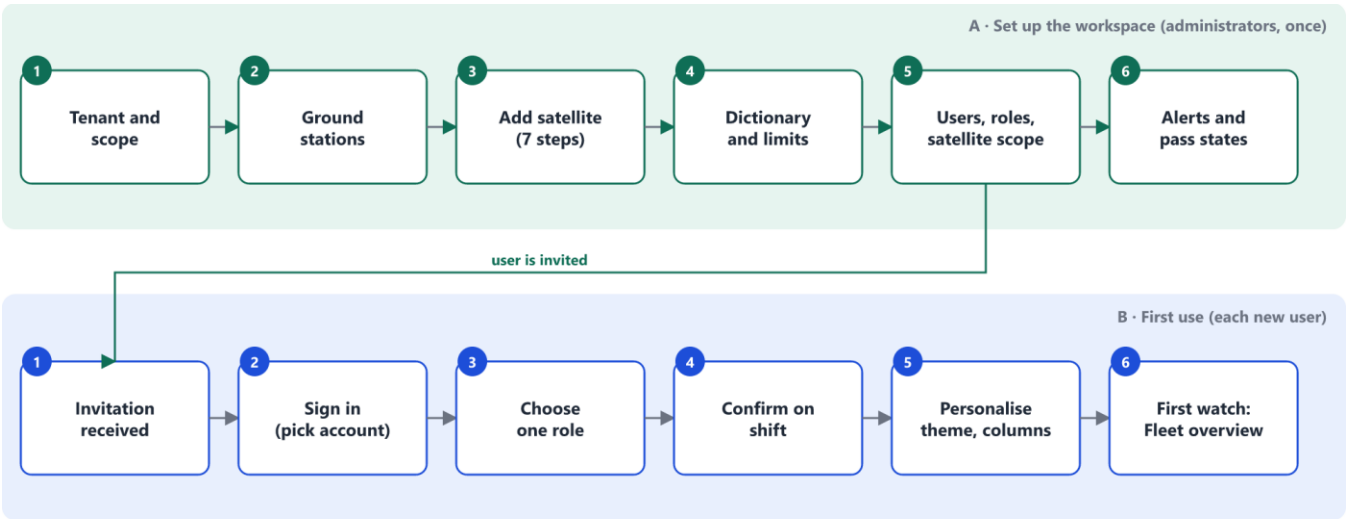


Figure 1. Onboarding: workspace set-up (top) and first use by each new user (bottom)

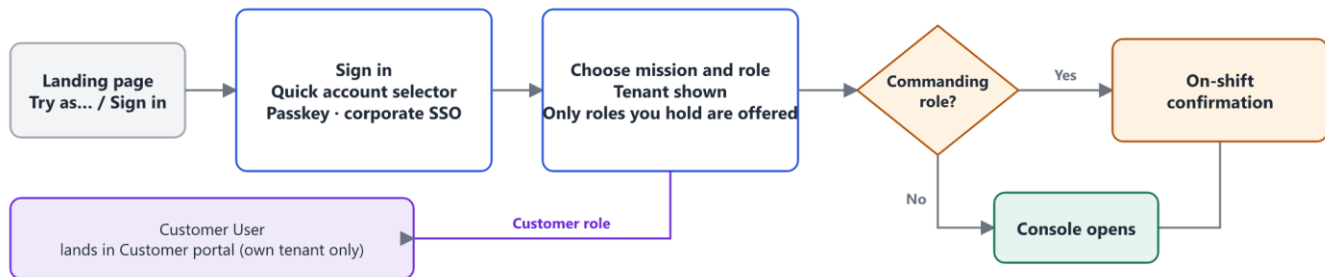
Set-up steps and where each is configured

#	Step	Who does it	Where in the console	What is set	Status
1	Tenant and scope	Platform Administrator	Users & access; role step at sign-in	Which tenant (Akashaveda, Nabhas Agritech, Terra Analytics) a person belongs to. One tenant never sees another's data.	Built
2	Ground stations	Ground Station Engineer	Ground stations	Provider, protocol (SLE, AWS Data/IP, own), bands, station health. Credentials live in the secrets store and are never displayed.	Built · Demo data
3	Add a satellite	Mission Planner or Flight Engineer	Fleet overview → Add satellite	Seven guided steps: register the spacecraft, declare virtual channels, load the telemetry dictionary, set alarm limits, correlate the on-board clock, book ground contacts, verify and enable.	Built · Demo data

#	Step	Who does it	Where in the console	What is set	Status
4	Dictionary and limits	Mission Database Engineer	Mission database	Upload and version the telemetry and command dictionary; each parameter carries warning and critical limits (low and high). Released versions are what the live decoder uses.	Built (JSON in backend; XTCE XML upload is Planned)
5	Users, roles, scope	Security Officer or Platform Administrator	Users & access → Invite operator	Person, one or more roles, satellite scope (for example AKV-* or NBH-01), passkey / MFA enforcement.	Built · Demo data
6	Alerting	Flight Director or Platform Administrator	Notifications & on-call	Routing rules, escalation policy and the on-call rota that Escalate hands an alarm to.	Built · Demo data
7	Pass states	Ground Station Engineer	Live pass monitor → Configure states	Name, description and the actions each pass state offers (section 5.3).	Built
8	Alarm thresholds	Flight Dynamics / Flight Director	Configuration file for now	Conjunction alarm distance (10 km, critical under 5 km) and screening window (4 h). Editing these from the console is Planned.	Planned

3. First sign-in and roles

A new user arrives at the landing page, either through Sign in or through the "Or jump in as" shortcuts. The role step offers only roles that person actually holds, and commanding roles need an on-shift confirmation before the console opens. A Customer User goes straight to the Customer portal and only ever sees their own tenant.



Role can be switched later from the avatar menu, but only to a role you hold. To act as someone else you sign in as them.

Figure 2. Sign-in, role choice and shift confirmation

In the demo, signing in means picking an account from the quick selector (grouped by role category). Paskey and corporate SSO buttons are present, but real identity is Planned.

Roles and what each sees

Category	Role	Screens	What the role is for
Operations	Spacecraft Operator	12	Monitor, acknowledge alarms, send routine commands, request critical ones.
	Flight Director	10	Approve critical commands, run recovery, escalation point.
Engineering	Flight Engineer	11	Analyse telemetry, author procedures, review advisories.
	Mission Database Engineer	2	Author, review and release dictionaries.
	ML Engineer	2	Anomaly models and advisory quality.

Category	Role	Screens	What the role is for
Planning & ground	Mission Planner	6	Build plans, book passes, manage deliveries.
	Ground Station Engineer	6	Operate stations and adapters, pass quality.
Governance & platform	Security Officer	3	Manage access, verify the audit ledger, break-glass.
	Platform Administrator	4	Platform health, deployments, on-call.
	System Administrator	all	Support and administration; no spacecraft authority.
Customers	Customer User	3	Own satellites, passes and deliveries. No commanding.

4. A working day in the console

Whatever the role, the day follows the same six stages. The table shows what the person does at each stage and where.

Stage	What the user does	Where	What they get
Orient	Check fleet health, who is in contact, what needs attention.	Fleet overview	Health counts, contact table, alarm bell.
Watch	Follow one satellite live; open several in their own windows.	Satellite health → Live	Cells, table or graphs; last and next contact; contact timeline.
Triage	Open an alarm, read its analysis, acknowledge, shelve or escalate.	Alarm console	Six alarm types, each with its own analysis view.
Act	Send a command or run a procedure; approve someone else's.	Command console, Approvals, Procedure runner	Safety checks, second-person approval, uplink status.
Plan	Book passes, set imaging, prepare the next contact.	Contact schedule, Mission plan, Live pass monitor	Planned procedures per contact; pass states.
Review	Look back: history, playback of a past pass, audit trail.	Satellite health → History / Playback; Audit ledger	Reconstructed history, replay, hash-chained record.

4.1 Fleet overview

- **Table of every satellite** with columns you choose. A Columns button lists 15 default columns (health, ground contact, last critical parameter, payload, last imaging, data downloaded and processed, procedure uplink %, conjunction, last and planned manoeuvre, last contingency) plus all 43 telemetry parameters. The choice is remembered; headers sort.
- **Globe and map** with layers you can switch on and off: satellites, orbit paths, labels, ground stations (each its own colour), coverage, place references, space debris, conjunction alerts. Satellites can be coloured by health, plane or individually, and the globe has seven map styles plus day/night, atmosphere and grid.
- **Observe one satellite:** hides all others and follows it with the camera, while zoom, tilt and layers keep working.
- **Conjunctions:** every 10 minutes each satellite is screened against a demo catalogue of debris and rocket bodies for the next 4 hours; close approaches are listed and critical ones raise a toast.

4.2 Satellite health

- **Live, History and Playback** are tabs on one page. History compares up to six parameters (overlay, stacked or side by side, zoom, pause and play, table and CSV export). Playback replays by time range, last duration, last pass or pass number, with play, speed and a scrubber.

- **Three views of the live values:** cells, a paged table, and a grid of small graphs. In each view an Edit button chooses which parameters appear; every view remembers its own choice.
- **Header shows only last and next contact.** A collapsible timeline below shows every contact 2 hours back and 10 hours ahead; hovering a segment shows AOS, LOS and the procedures and commands planned on it, with the percentage completed.
- **Side panel:** open alarms, payload and data status, procedure uplink progress, last events (imaging, dump, manoeuvre), what is planned on board, flight-dynamics summary and conjunction status.
- **New window** opens the satellite alone in its own browser window, so several can be watched side by side.

5. Key flows

5.1 Alarm handling

Alarms come in six types: health (limit violations, grouped by subsystem), conjunction, payload, commands and procedures, ground contact, and data processing. Clicking any alarm opens an analysis that fits its type: limits and a recent chart for health, encounter geometry and recommended steps for a conjunction, station status and affected contacts for a ground alarm, and so on.

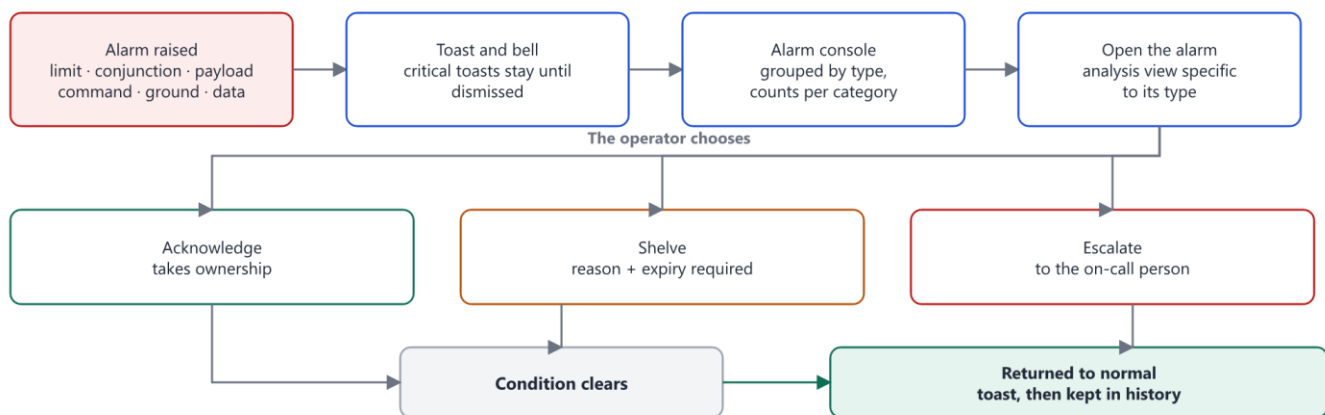


Figure 3. Alarm lifecycle from raised to returned-to-normal

5.2 Sending a critical command

Routine commands go straight through the safety checks. A critical command adds a second person: the requester cannot approve their own request, and the approver confirms with a fresh passkey touch after seeing the interlock telemetry captured at request time. A command is only marked completed when telemetry shows its effect, not merely because the spacecraft acknowledged it.

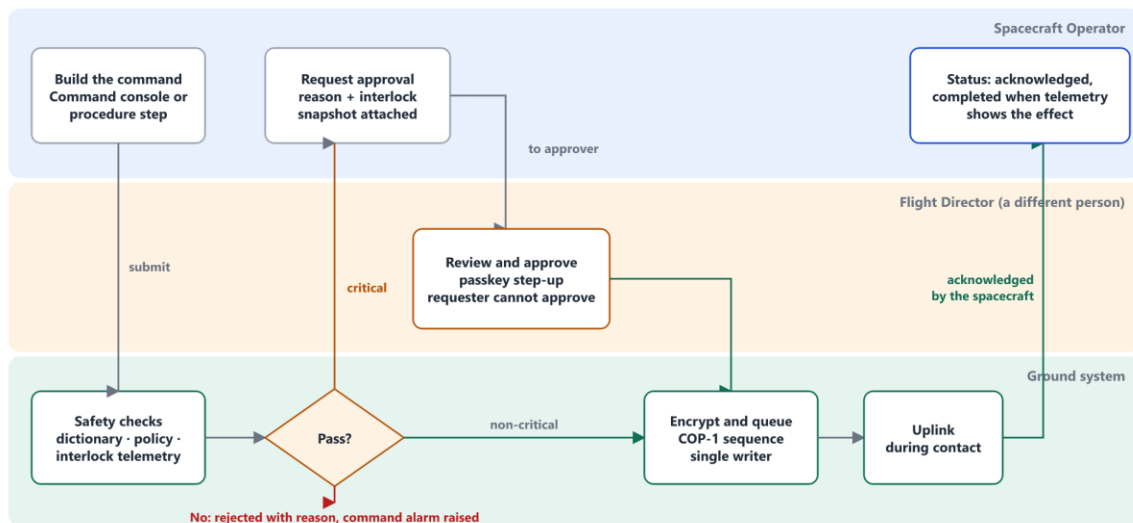
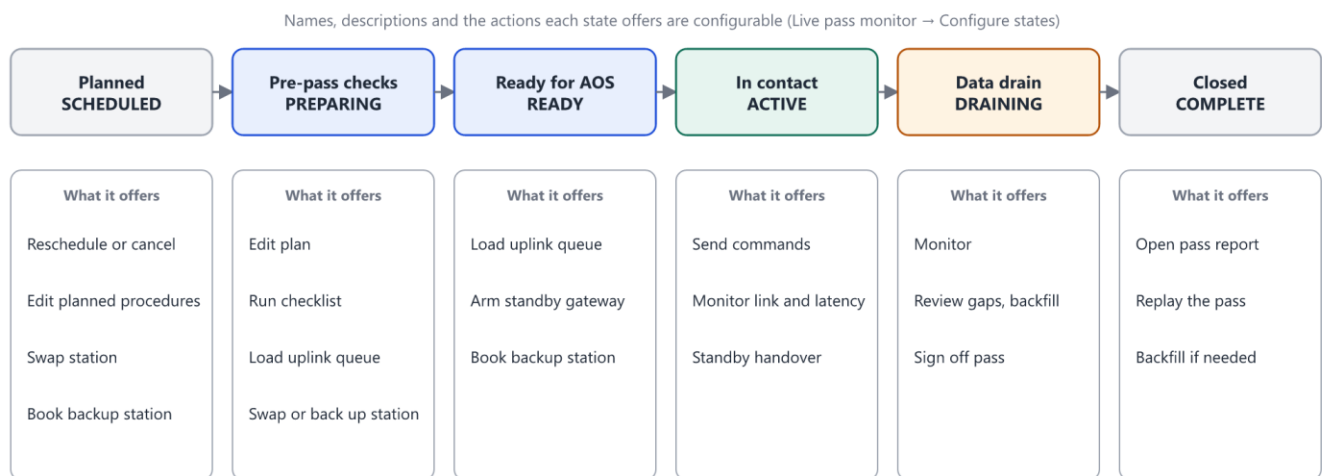


Figure 4. Critical command: request, approval, uplink and confirmation

5.3 A contact from plan to sign-off

Each pass moves through six states. The default names below are suggestions and can be renamed; the description and the actions each state offers are also configurable from the Live pass monitor. The monitor shows the current state's actions as buttons.



AOS = satellite rises above the horizon · LOS = it sets · Defaults shown

Figure 5. Pass states and what each offers (defaults)

6. Configuration and personal settings

Some settings belong to the whole workspace, others to the person. Personal settings are stored in the browser today, so they follow the browser, not the account; moving them to the account is Planned.

Setting	Where to change it	Applies to	Default
Theme (light / dark)	Sun / moon button in the top bar and on sign-in	Person (browser)	Dark
Fleet table columns	Fleet overview → Columns	Person (browser)	15 columns
Satellite health view and which parameters show	Satellite health → view switch and Edit	Person (browser)	Cells, all parameters

Setting	Where to change it	Applies to	Default
Globe layers, colour mode, map style	Globe → Layers button	Person (browser)	Satellites, orbits, stations, places, conjunctions
Active role	Avatar menu (only roles you hold)	Session	First role held
Pass state names, descriptions, actions	Live pass monitor → Configure states	Browser today; workspace Planned	Planned, Pre-pass checks, Ready for AOS, In contact, Data drain, Closed
Dictionary, limits, satellites, stations	Mission database, Add satellite, Ground stations	Workspace	Per satellite
Users, roles, satellite scope, MFA	Users & access	Workspace	None until invited
Alarm routing and escalation	Notifications & on-call	Workspace	Rota and rules as defined

7. One journey per persona

Persona	From invitation to first useful day
Spacecraft Operator	Invited by the Security Officer with role and satellite scope. Signs in, picks the Operator role, confirms on shift. Fleet overview shows who is in contact. Opens AKV-03 in its own window and watches the graphs view. A heater alarm arrives: the toast opens the Alarm console, the analysis shows the limit and trend, the operator acknowledges and starts the recovery procedure. The critical step needs approval, so the operator requests it and waits for the Flight Director; status follows the command to completed.
Flight Director	Signs in with the Flight Director role and confirms on shift. The bell and Approvals show pending requests. Reviews who asked, why, and the interlock snapshot, approves with a passkey touch. Handles escalations from the alarm console and owns conjunction decisions: reads the encounter analysis and requests a manoeuvre plan.
Mission Planner	Signs in, opens Contact schedule and the Live pass monitor. Books passes and edits planned procedures per contact. Onboards a new satellite through the seven-step wizard, then checks it appears in the Fleet overview table. Watches pass states move from Planned to Closed and opens the pass report.
Customer User	Invited with a tenant scope only. Signs in and lands in the Customer portal. Sees own satellites, upcoming passes, deliveries and tasking requests, and manages API keys and webhooks. No commanding, no other tenant's data.

8. What is real today

- **Real:** the telemetry path from simulator through decoding to the console, limit alarms, the command chain with safety checks and COP-1 acknowledgement, contact windows and satellite positions from an orbit model, and conjunction screening geometry.
- **Demo data:** payload, data-processing, procedure and flight-dynamics figures, the debris catalogue, history for parameters (reconstructed around the live value), and most administration screens.
- **Planned:** real sign-in (SSO, passkeys), personal settings stored on the account, XTCE upload, editable alarm thresholds, live tracking data for conjunctions, and separate windows sharing one live session.